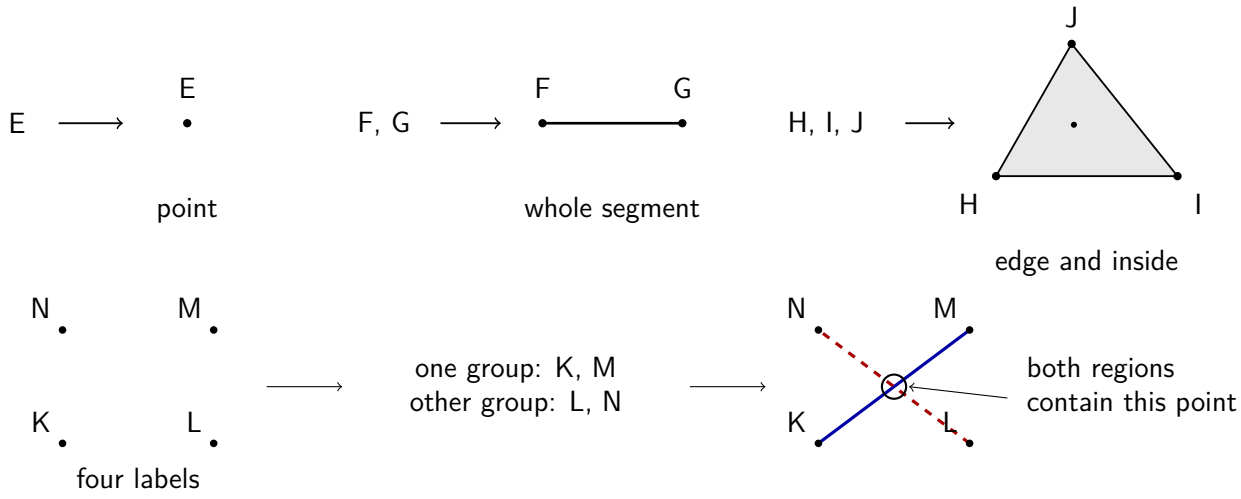
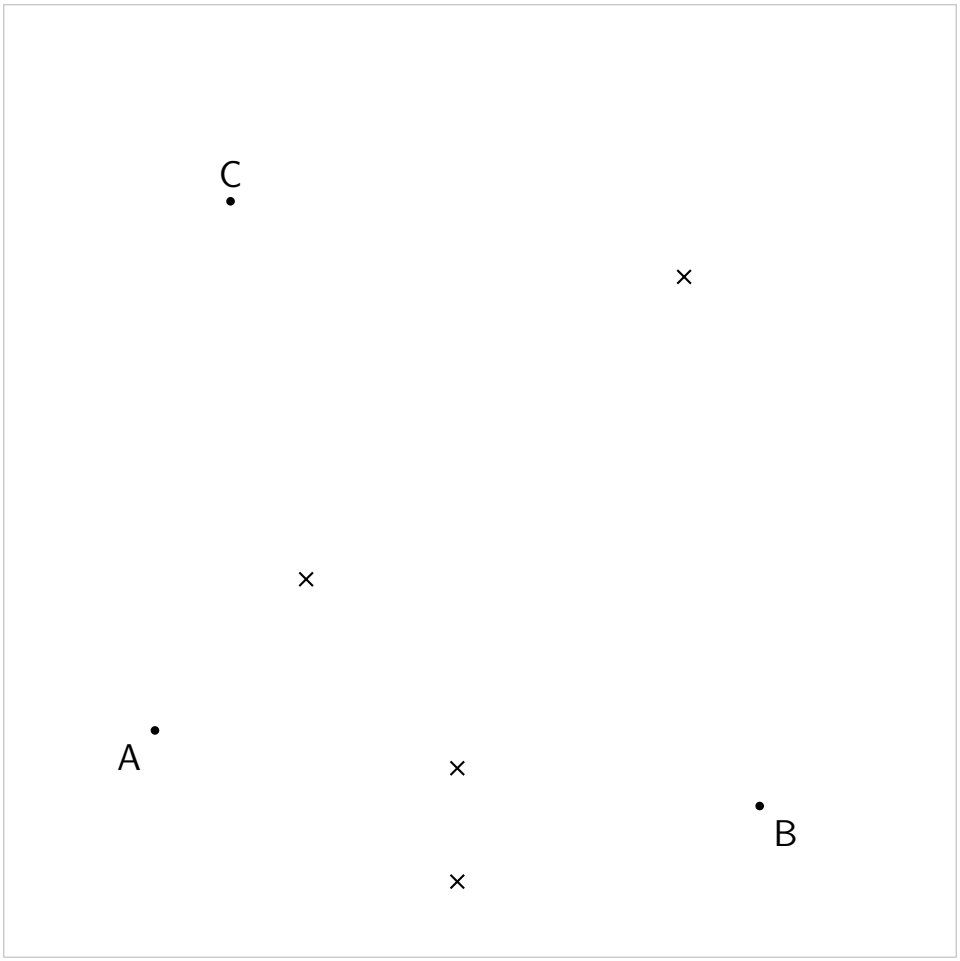
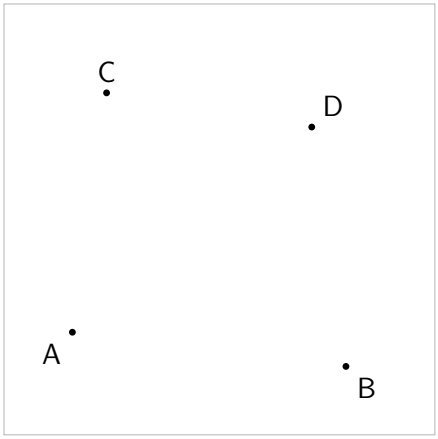
Use each label once in two groups, with at least one label in each. Draw each group's stretched-band region and lightly shade its inside. Use the centers of the marks. Sharing even one point counts. Swapping the two groups is the same split.

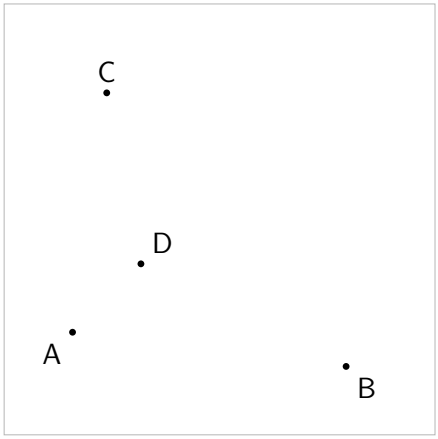


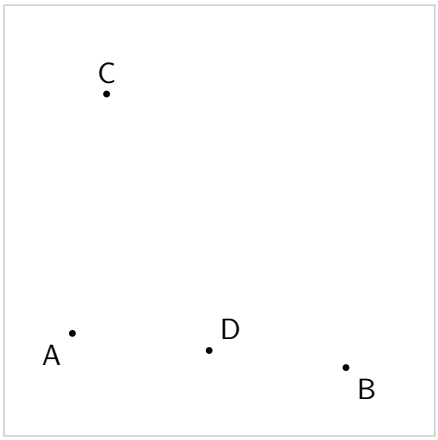
Problem 1: Put D on each cross. For each position, find a split of A, B, C, and D whose two regions meet.

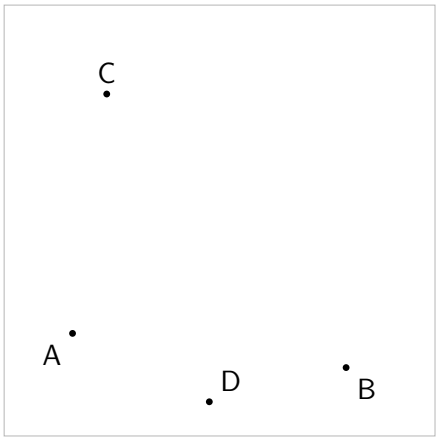


Problem 1: For each position of D, draw the two regions of your split and mark a shared point.

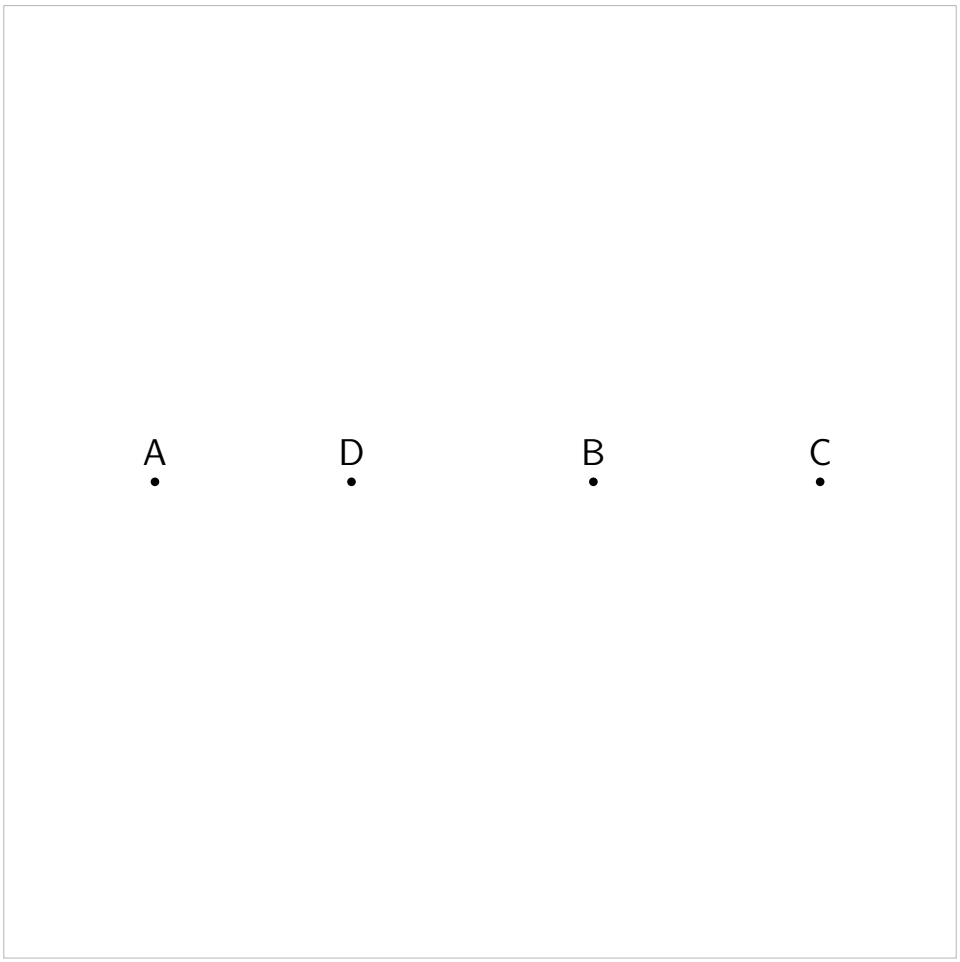




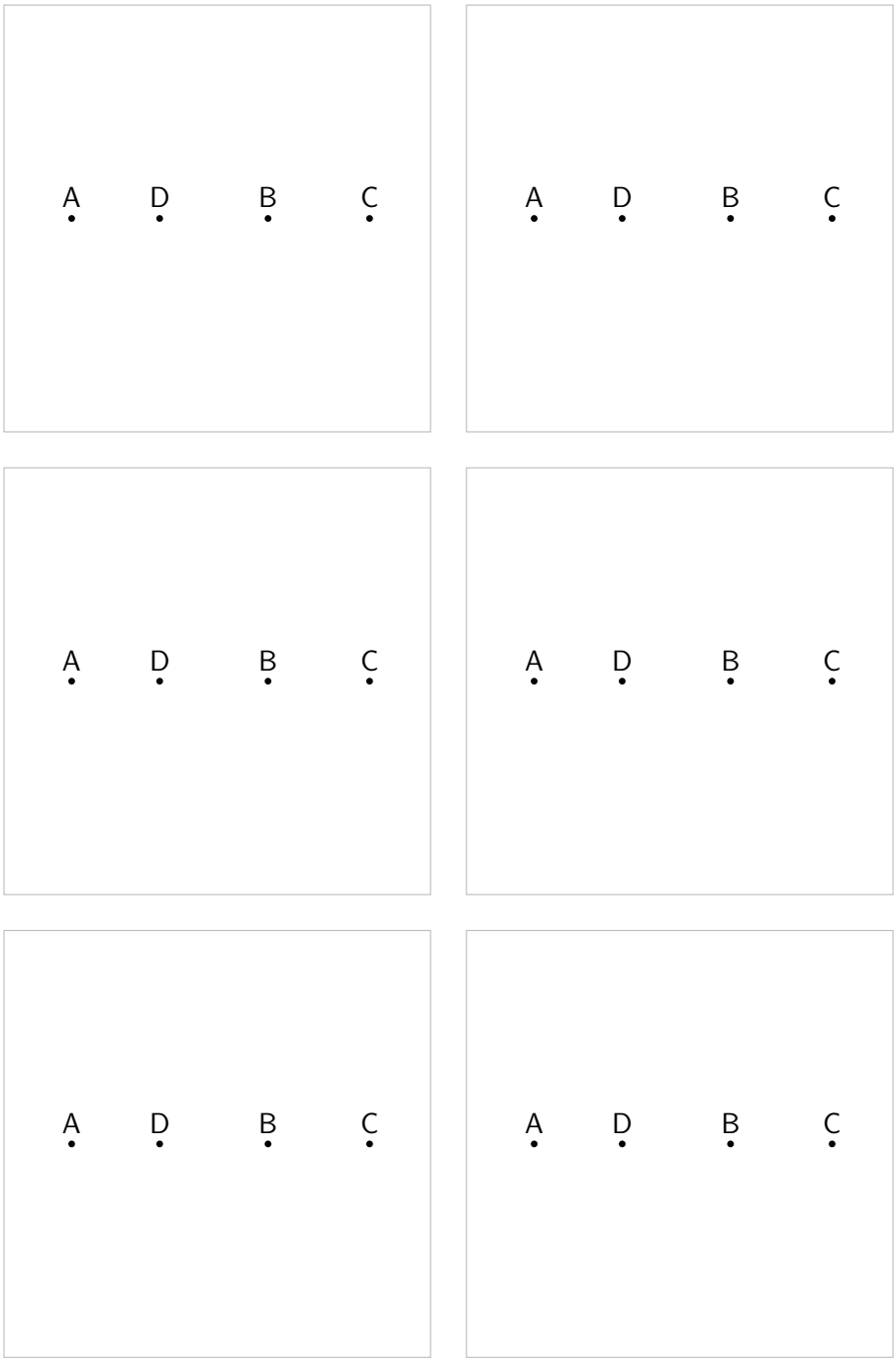




Problem 2: Find every split of A, B, C, and D whose regions meet. Explain how you know your list is complete.



Problem 2: Find every split of A, B, C, and D whose regions meet. Explain how you know your list is complete.

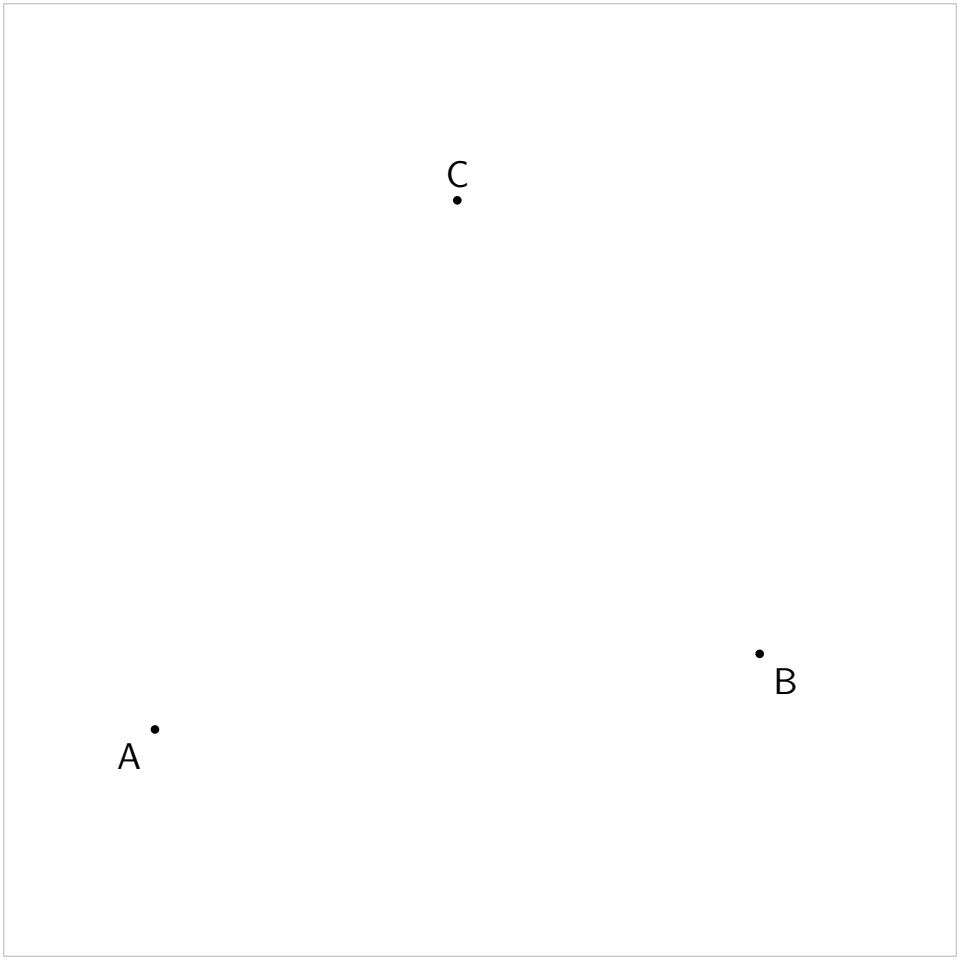


Problem 3: Place A, B, C, and D so that their two regions share a whole straight segment. Can two groups made from these four labels share a filled triangle? Explain.

Problem 4: Make two arrangements of A, B, C, and D. No split may work for both arrangements. Show a working split for each.

Problem 4: Make two arrangements of A, B, C, and D. No split may work for both arrangements. Show a working split for each.

Problem 5: Can these three dots be split into two groups whose regions meet? Explain why your answer covers every split. Keep A and B fixed. Find every place in the box where C could go so that the regions meet.



Problem 6: Take turns placing A, B, C, and D in the box. Try to make an arrangement that the other player cannot split successfully. Can four dots always be split so their regions meet?